Shantanu Singh

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EDUCATION

Bachelor of Technology (B-Tech.) | IIT Patna — **Bihta, Bihar**
Metallurgical and Materials Engineering, CPI: 8.20/10.00 — *July 2020 – June 2024*

EXPERIENCE

Tarento Technologies — **Mar 2025 – Present**
Software Engineer, — *Bangalore, Karnataka*

- Contributing to the development of an agentic AI system using **Python**, **LangChain**, **LangGraph**, **AutoGen**, and **MCP servers** for personalized learning diagnostics.
- Designed modular pipelines for extracting mathematical concepts from Class 6 NCERT textbooks, maintaining structured outputs using **Pydantic models**.
- Applied **prompt engineering** techniques to fine-tune LLM behavior and improve concept extraction and question generation accuracy.
- Utilized **LangSmith** to debug and analyze **LLM traces**, optimizing agent workflows and improving system reliability.
- Built and maintained a **FastAPI**-based backend with well-structured routing; analyzed student and concept data using **PostgreSQL**.
- Managed infrastructure using **Docker**, **Redis**, **TimescaleDB**, **Neo4j**, and **PostgreSQL**, enabling graph-based storage and synchronization of concept relationships.
- Integrated and tested **CrewAI agents** as simulated students to validate diagnostic question generation workflows.

Futures First — **July 2024 - Mar 2025**
Market Analyst, — *Kolkata, West Bengal*

- Traded multiple asset classes on global exchanges, including **ICE Europe(Brent)** and **NYMEX US(WTI)**, leveraging Fundamental, Technical, and **Quantitative analysis** to develop high-impact trading strategies.
- Collaborated with **cross-functional teams** to design **APIs** for seamless data exchange between trading platforms, improving the system's interoperability.
- Developed and **implemented trading algorithms** to automate strategies, improving **efficiency** and profitability.
- Conducted data analysis to identify patterns, enabling data-driven decision-making.

Tailwnd — **Feb 2023 - Aug 2023**
Software Development Intern, — *Gurgaon, New Delhi*

- Developed and improved features for a concept mapping system, focusing on scalability, performance optimization and debugging.
- Automated deployment pipelines using CI/CD practices, ensuring seamless integration and delivery.
- Collaborated in an agile team to implement reusable components with React.js and JavaScript.
- Designed and implemented **RESTful APIs** to enable seamless data retrieval and integration between components, improving back-end efficiency and scalability.
- Conducted **API testing and debugging** to ensure robust functionality, reliability, and adherence to industry best practices.

PROJECTS

M-Tech Portal,IIT Patna — **Website Link**
ReactJs,NodeJs,MongoDB,AWS, — *March 2022*

- Built a full-stack portal using React.js, Node.js, and MongoDB, deployed on AWS to streamline the M-Tech admissions process.
- Successfully streamlined the admissions process by **enabling 1000+ students** to apply to multiple branches at once through the portal, **reducing admission time by 50 percent.**

Transcript Generator — **Source Code**
Python, FPDF2, StreamLit, — *November 2022*

- **Objective:**To build an **app** to generate transcript in PDF format from raw **CSV** data.Raw data included details of students, courses, grades and results.
- The content placement was determined dynamically, based on the number of semesters and courses.**FPDF2** was used for generating and filling the pdf and **Streamlit** was used to create the frontend.

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Java, SQL, HTML, JavaScript
Frameworks: LangChain, LangGraph, AutoGen, CrewAI, FastAPI, React.js, Node.js, Pandas, Pydantic, Microservices
Tools & Platforms: LangSmith, MCP Servers, Google ADK, Cursor IDE, Linux, Docker, Kubernetes, Jenkins, Terraform, CI/CD, Neo4j, PostgreSQL, TimescaleDB, Redis
Professional Skills: Data Structures and Algorithms, Object-Oriented Programming (OOP), Graph Theory, Number Theory, Operating Systems, RDBMS, API Testing, TCP/IP, Networking (VPC, Load Balancers, TLS), Security Best Practices
Soft Skills: Leadership, Team Management, Public Speaking, Analytical Thinking, Problem Solving, Decision-Making

CREDENTIALS AND ACHIEVEMENTS

CodeKaze June'23 (Coding Ninjas): Achieved **AIR 2442 out of 137,117 participants** after clearing two rounds including Aptitude based test and a Coding Round.
JumpStart August'22 : Participated in **Jumpstart hackathon** conducted by **Publicis Sapients** and selected in **top 400 out of 30,000 candidates** after qualifying two Coding rounds.
Solved more than **1200 problems** on various platforms like **Leetcode**, GeeksforGeeks, Codeforces, and CSES.